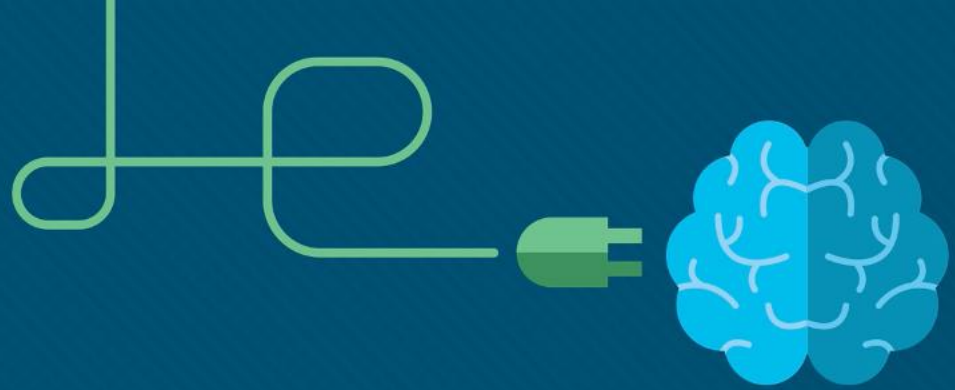




Module 2: Formatting Text and Organizing Data

Introduction to HTML



Module Objectives

Module Title: Introduction to HTML

Module Objective: This module dives into the essentials of text formatting and data organization in HTML. You'll work with headings, paragraphs, text formatting options, lists, and tables to create structured and readable content for your web pages.

Skills you'll gain: Mastery of text formatting, structuring content with lists and tables, and creating organized and accessible web layouts.

At the end of this module, you'll complete a project that consolidates your knowledge and a test to evaluate your progress.

Module Objectives

Course Syllabus:

The course is organized into six modules, each focusing on key aspects of HTML, along with a final project and test to solidify your learning. Below is a detailed breakdown of the syllabus:

Module 2: Formatting Text and Organizing Data

- Headings
- Paragraphs and Text Formatting – Part 1
- Paragraphs and Text Formatting – Part 2
- Paragraphs and Text Formatting – Part 3
- Paragraphs and Text Formatting – Part 4
- Lists
- Tables

2.1 Headings

2.1.1 Intro to Headings

HTML headings are special elements in an HTML document that are used to define the heading or title of a section or page. Headings provide structure and hierarchy to a web page by indicating which sections are more important or subordinate to others.

There are six levels of HTML headings, ranging from `<h1>` to `<h6>`, with `<h1>` being the most important heading and `<h6>` being the least important. The heading level is determined by the number in the heading tag. For example, `<h1>` indicates the most important heading, while `<h2>` indicates a subheading under `<h1>`, and so on.

2.1.2 The Purpose of Headings

The primary purpose of headings is to provide accessibility and improve the user experience by helping visitors understand the content and structure of a web page.

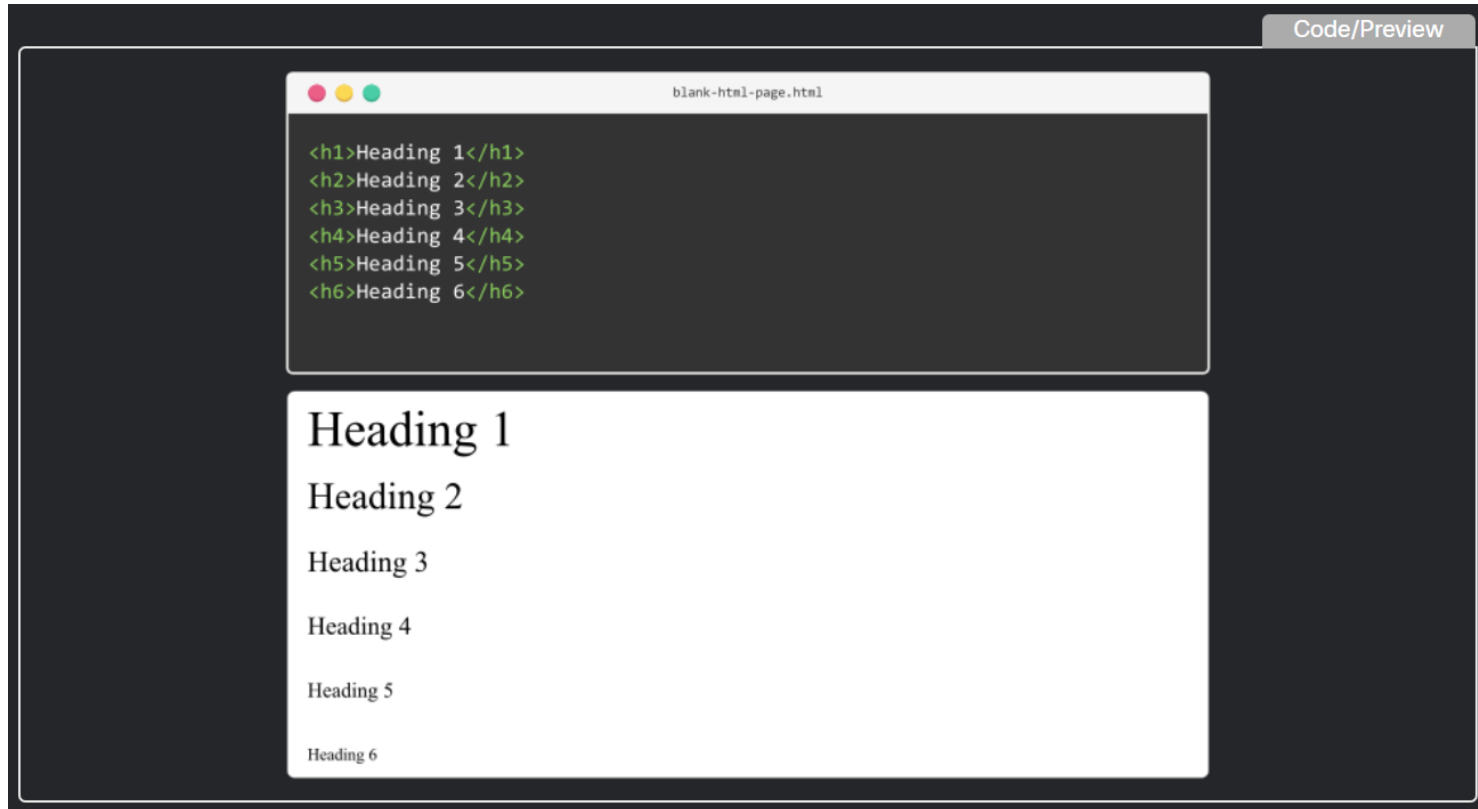
Headings are also important for search engines as they use them to understand the topic of a web page and index1 (categorize) it accordingly. Therefore, proper use of HTML headings can improve the search engine optimization2 (SEO) of a web page by making it easier for search engines to identify and rank relevant content.

It's important to use headings appropriately and not misuse them for stylistic purposes. The content of the headings should accurately reflect the content they precede, and they should not be skipped or misused to create a visual effect.

For example, don't use `<h3>` as a subheading for an `<h1>` without including an `<h2>` in between. This can be confusing for users and search engines.

2.1.2 The Purpose of Headings

Here is an example of how to use headings in HTML5:



The screenshot displays a code editor window titled "blank-html-page.html" with a "Code/Preview" toggle. The code editor shows the following HTML5 heading tags:

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```

The preview window shows the rendered output of these headings, demonstrating their visual hierarchy:

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

2.1.3 Heading Sizes

In HTML, the size of headings is defined by the HTML tags `<h1>` to `<h6>`, but the actual display size can vary depending on the browser and the device used to view the page. By default, the font size of headings is determined by the browser's default stylesheet or user-defined stylesheets.

Typically, the default font size for the different heading levels in most web browsers is as follows:

`<h1>`: 32px

`<h2>`: 24px

`<h3>`: 18px

`<h4>`: 16px

`<h5>`: 14px

`<h6>`: 12px

While the size of the heading can be adjusted using CSS, it's generally best to use the default sizes provided by HTML5, as this ensures that the headings are consistent across different browsers and devices.

2.1.4 Heading Levels

The content that can be included between headings can include any text or other HTML elements that are relevant to the content on the page. For example, an `<h1>` heading might be used for the title of the page, while `<h2>` headings might be used for major sections of the page, with relevant text or other HTML elements included between each of the headings to provide more specific information.

Here is a breakdown of different levels of HTML5 headings:

`<h1>`: The largest and most important heading on the page. This is typically used for the title of the page or the main heading.

`<h2>`: A subheading of `<h1>`. This is often used for the main section headings on a page.

`<h3>`: A subheading of `<h2>`. This is often used for subsection headings or less important main headings.

`<h4>`: A subheading of `<h3>`. This is often used for subsection headings or less important main headings.

`<h5>`: A subheading of `<h4>`. This is often used for subsection headings or less important main headings.

`<h6>`: The smallest and least important heading on the page. This is often used for minor headings or subheadings.

2.2 Paragraphs and Text Formatting – Part 1

2.2.1 The <p> Tag and Inline Formatting

In HTML, paragraphs are defined using the <p> tag.

The <p> tag indicates that the enclosed text should be treated as a separate paragraph, with a blank line inserted above and below the text. The <p> tag is a block-level element, which means that it takes up the entire width of the container¹ in which it is placed.

To create a paragraph, you can use the opening <p> tag before the text and the closing </p> tag after the text that you want to display as a paragraph.



2.2.2 Formatting Text with Inline Tags

HTML provides several inline tags that can be used to format text within a paragraph. Here are some of the most common tags that can be used for formatting text within a paragraph:

The Tag

The tag is used to make the enclosed text **bold**. For example, to make the word "important" bold in a paragraph, you would write:

```
<p>This is some <strong>important</strong> text.</p>
```

The Tag

The tag is used to make the enclosed text **italic**. For example, to make the word "emphasis" italic in a paragraph, you would write:

```
<p>This is some text that requires <em>emphasis</em>.</p>
```

2.2.2 Formatting Text with Inline Tags

HTML provides several inline tags that can be used to format text within a paragraph. Here are some of the most common tags that can be used for formatting text within a paragraph:

The <u> Tag

The <u> tag is used to underline the enclosed text. For example, to underline the word "underlined" in a paragraph, you would write:

```
<p>This is some text that needs to be <u>underlined</u>.</p>
```

index.html



The <sup> Tag

The <sup> tag is used to make the enclosed text superscript (slightly above the regular text). For example, to write ordinal numbers, such as first, second, and third in the following form: "1st, 2nd, and 3rd", you would write:

```
<p>1<sup>st</sup>, 2<sup>nd</sup>, and 3<sup>rd</sup></p>
```

index.html



2.2.2 Formatting Text with Inline Tags

HTML provides several inline tags that can be used to format text within a paragraph. Here are some of the most common tags that can be used for formatting text within a paragraph:

The <sub> Tag

The <sub> tag is used to make the enclosed text subscript (slightly below the regular text). For example, to write a chemical formula for water, you would write:

```
<p>The chemical formula for water is H<sub>2</sub>O.</p>
```

index.html



The Tag

The tag is used to strike through text, indicating that it has been deleted or is no longer relevant. For example:

```
<p>Special Price: <del>$100</del> $75.</p>
```

index.html

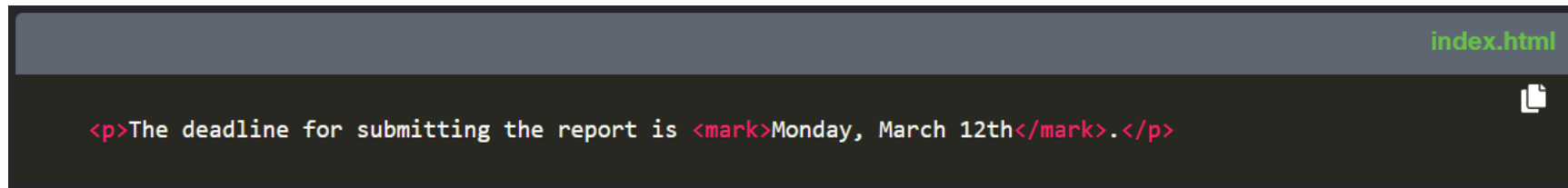


2.2.2 Formatting Text with Inline Tags

HTML provides several inline tags that can be used to format text within a paragraph. Here are some of the most common tags that can be used for formatting text within a paragraph:

The <mark> Tag

The <mark> tag is used to highlight text with a background color, indicating that it is of special importance. For example, to highlight the date "Monday, March 12th", and draw attention to the deadline, you would wrap it within the <mark> tag:

A screenshot of a code editor with a dark theme. The top bar is dark gray with the text 'index.html' in green on the right. The main editor area is black and contains the following HTML code in a light blue font:

```
<p>The deadline for submitting the report is <mark>Monday, March 12th</mark>.</p>
```

 A small white icon of a document is visible in the bottom right corner of the editor area.

2.3 Paragraphs and Text Formatting – Part 2

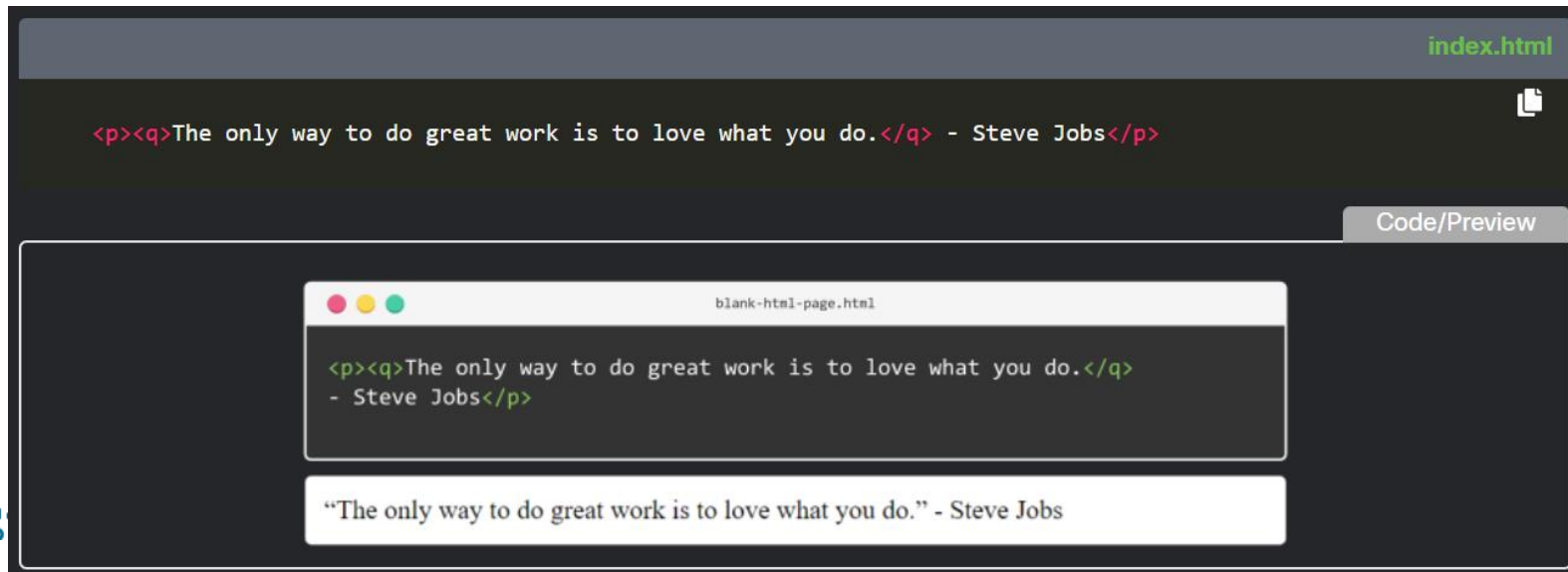
2.3.1 Quotations: the <q> and <blockquote> Tags

When you want to add a quote to your HTML document, you can use two tags:

- the <q> tag, and
- the <blockquote> tag.

The <q> Tag

The <q> tag is used for short, inline quotes (typically just a few words or a short phrase). It is often rendered with quotation marks around the text. For example:



2.3.1 Quotations: the <q> and <blockquote> Tags

The <blockquote> Tag

The <blockquote> tag is used for longer, block quotes. It is often used to quote entire paragraphs or sections of text, and is typically rendered with indentation or a different font to set it apart from the surrounding text. For example:

Code/Preview

blank-html-page.html

```
<p>As Albert Einstein once said:</p>
<blockquote>
  <p>"Imagination is more important than knowledge. Knowledge is limited.
Imagination encircles the world. The true sign of intelligence is
not knowledge but imagination."</p>
</blockquote>
<p>This quote speaks to the idea that creativity and original thinking
are more important than simply accumulating knowledge.
It suggests that being able to see beyond what is already knownand
to imagine new possibilities is a key characteristic of intelligence.</p>
```

As Albert Einstein once said:

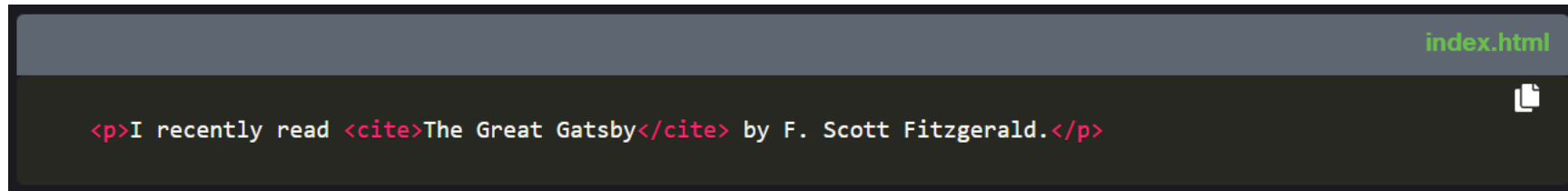
"Imagination is more important than knowledge. Knowledge is limited. Imagination encircles the world. The true sign of intelligence is not knowledge but imagination."

This quote speaks to the idea that creativity and original thinking are more important than simply accumulating knowledge. It suggests that being able to see beyond what is already known and to imagine new possibilities is a key characteristic of intelligence.

2.3.2 The <cite> Tag

When you want to mark up the title or source of a book, film, song, or any other creative work, provide attribution or credit for an idea or information, or indicate a reference to another source, you can use the <cite> element.

For example:



```
index.html

<p>I recently read <cite>The Great Gatsby</cite> by F. Scott Fitzgerald.</p>
```

2.3.3 The <cite> Attribute

When using quotes, it's important to attribute the source of the quote, typically by providing a URL. One way to do this is with the <cite> attribute, which you can use inside a <q> tag to specify the source of the quote.

For example:

```
<p>As Steve Jobs once said,  
<q cite="https://www.brainyquote.com/quotes/steve_jobs_416096"> Innovation  
distinguishes between a leader and a follower.</q></p>
```

index.html

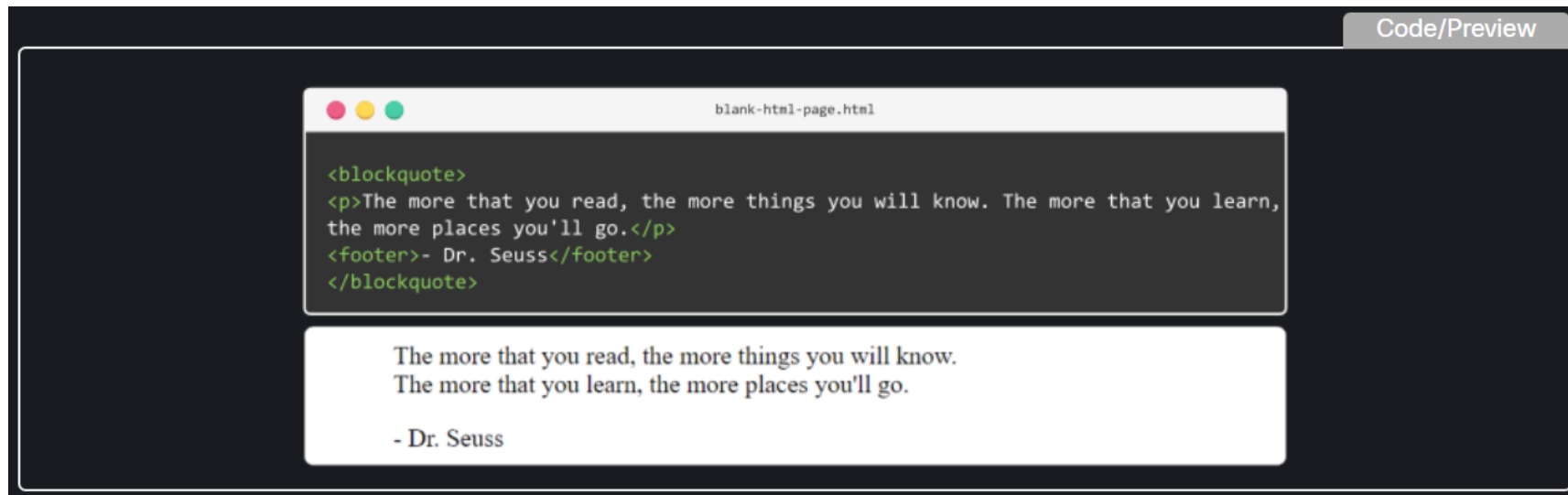


```
<p>As Steve Jobs once said, <q cite="https://www.brainyquote.com/quotes/steve_jobs_416096">Innovation distinguis
```

2.3.4 The <footer> Tag

Another way to provide more context for a quotation is to use the <footer> tag, which is a semantic tag that helps search engines and screen readers understand the content of the page.

For example:



2.4 Paragraphs and Text Formatting – Part 3

2.4.1 Line Breaks and Horizontal Rules

In HTML, a line break is a simple element that creates a new line of text within a block-level element. This is typically represented by the `<br>` tag and is used to separate lines of text, such as in poetry or song lyrics.

To add a line break, insert the `<br>` tag wherever you want to create a new line of text. It's important to note that the `<br>` tag is an empty element, meaning it doesn't require a closing tag.

For example, the following code creates two lines of text with a line break between them:



2.4.1 Line Breaks and Horizontal Rules

A horizontal rule is a visual element that creates a horizontal line across the width of a block-level element. It is typically used to separate sections of content within a page.

To add a horizontal rule, insert the `<hr>` tag wherever you want to add a line. The `<hr>` element is self-closing and does not require a closing tag.

For example, the following code creates a horizontal rule:

Code/Preview

blank-html-page.html

```
<p>This is some text above the line.</p>
<hr>
<p>This is some text below the line.</p>
```

This is some text above the line.

This is some text below the line.

2.5 Paragraphs and Text Formatting – Part 4

2.5.1 Computer Code and Preformatted Text

In HTML, the `<code>` and `<pre>` tags are used to display computer code and preformatted text, respectively.

These tags are useful when displaying code examples or preserving the formatting of text that has been entered with spaces or line breaks.

The `<code>` tag is used to display inline code snippets.

When used, the text inside the `<code>` tag is displayed in a monospaced font, which makes it stand out from the surrounding text. This is useful when you want to display code in the middle of a paragraph, like this:

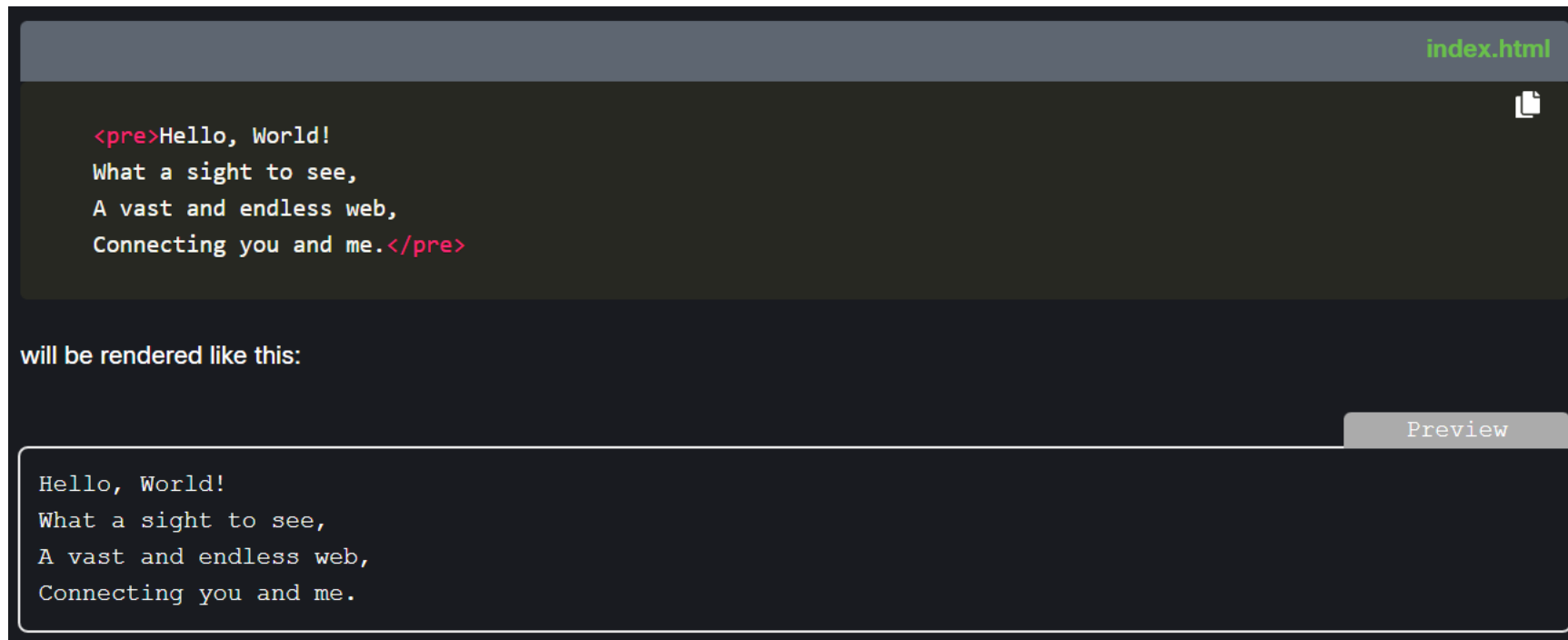
Preview

Hello, World!

If you want to display a code snippet in a block, you'll need to use the `<pre>` tag instead.

2.5.2 Displaying preformatted text with the <pre> tag

To display preformatted text like computer code, poetry, or song lyrics in HTML, you can use the <pre> tag. For example, the following formatted text:



The screenshot shows a code editor interface. At the top right, a tab is labeled "index.html". The main editor area contains the following HTML code:

```
<pre>Hello, World!  
What a sight to see,  
A vast and endless web,  
Connecting you and me.</pre>
```

Below the code editor, the text "will be rendered like this:" is displayed. To the right of this text is a "Preview" button. The preview area shows the rendered output of the code:

```
Hello, World!  
What a sight to see,  
A vast and endless web,  
Connecting you and me.
```

2.6.1 What are HTML Lists

Lists are a fundamental part of HTML and are used to present information in an organized and structured manner. They allow you to group a set of related items or pieces of information together.

Whether you're creating a to-do list, a shopping list, a recipe, or a list of fun facts, HTML lists have got you covered!

There are three types of lists in HTML:

- unordered lists,
- ordered lists, and
- definition lists.

2.6.2 Unordered Lists

Unordered lists are created with the `<ul>` tag, and are used to display a list of items with no particular order. Each item in the list is represented by the `<li>` tag.

For example:

Code/Preview

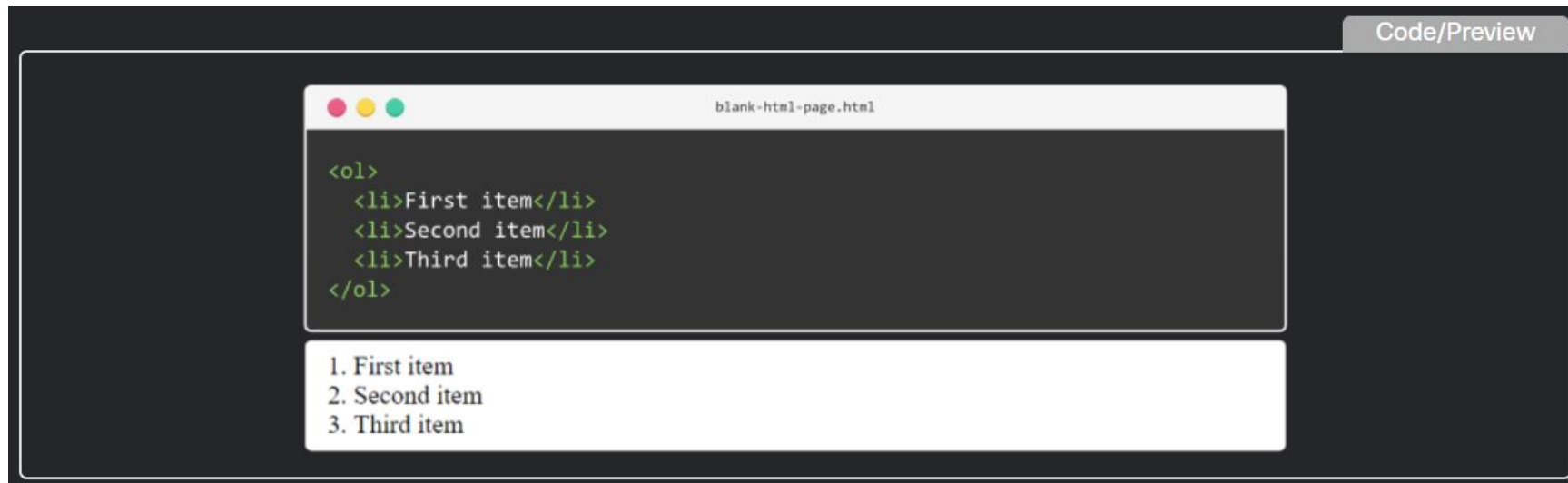


```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

- Item 1
- Item 2
- Item 3

2.6.3 Ordered Lists

Ordered lists are created with the `<ol>` tag, and are used to display a list of items in a specific order. Just like in the case of unordered lists, each item in an ordered list is represented by the `<li>` tag. For example:



2.6.4 Definition Lists

Definition lists are created with the `<dl>` tag, and are used to display a list of terms and their definitions. Each term is represented by the `<dt>` tag, and each definition is represented by the `<dd>` tag.

For example:

Code/Preview

blank-html-page.html

```
<dl>
  <dt>Term 1</dt>
  <dd>Definition 1</dd>
  <dt>Term 2</dt>
  <dd>Definition 2</dd>
  <dt>Term 3</dt>
  <dd>Definition 3</dd>
</dl>
```

Term 1
Definition 1
Term 2
Definition 2
Term 3
Definition 3

2.6.5 Nested Lists

HTML allows you to nest lists within other lists. This means you can create a hierarchy of information, where sub-lists are indented to show they belong to a parent list item.

To create a nested list, simply place a new `<ul>` or `<ol>` element within an existing `<li>` element.

Code/Preview

blank-html-page.html

```
<ul>
  <li>First item</li>
  <li>Second item
    <ul>
      <li>Sub-item 1</li>
      <li>Sub-item 2</li>
    </ul>
  </li>
  <li>Third item</li>
</ul>
```

- First item
- Second item
 - Sub-item 1
 - Sub-item 2
- Third item

2.6 Tables

2.7.1 What are HTML Tables

HTML Tables are used to display data in rows and columns. They are a great way to present data in a structured and organized way.

Tables are made up of rows and columns, and each intersection of a row and a column is called a cell. Each cell in a table can contain text, images, or other HTML elements.

2.7.2 Creating a Basic Table

To create a basic table, you need to use three elements: `<table>`, `<tr>`, and `<td>`.

The `<table>` element defines the entire table, and the `<tr>` element defines a row within the table. The `<td>` element defines a cell within a row.

Here's an example of a basic table with two rows and two columns:

Code/Preview

blank-html-page.html

```
<table>
  <tr>
    <td>Row 1, Column 1</td>
    <td>Row 1, Column 2</td>
  </tr>
  <tr>
    <td>Row 2, Column 1</td>
    <td>Row 2, Column 2</td>
  </tr>
</table>
```

Row 1, Column 1 Row 1, Column 2
Row 2, Column 1 Row 2, Column 2

2.7.3 Adding Headers

You can also add header rows to your table using the `<th>` element. The `<th>` element works the same as the `<td>` element, but it is used to define header cells. Header cells are often displayed in bold and centered by default. Here's an example of a table with a header row:

Code/Preview

blank-html-page.html

```
<table>
<tr>
  <th>Service</th>
  <th>Price</th>
</tr>
<tr>
  <td>Web design</td>
  <td>$500</td>
</tr>
<tr>
  <td>Graphic design</td>
  <td>$300</td>
</tr>
<tr>
  <td>Content writing</td>
  <td>$200</td>
</tr>
<tr>
  <td>Social media management</td>
  <td>$150</td>
</tr>
</table>
```

Service	Price
Web design	\$500
Graphic design	\$300
Content writing	\$200
Social media management	\$150

2.7.4 Merging Cells

In HTML tables, it is possible to merge two or more cells in a row or a column to create a larger cell that spans multiple rows or columns. This can be useful for creating headings, subheadings, or data cells that have to span across several columns or rows.

You can merge cells in a table using the `colspan` and `rowspan` attributes. The `colspan` attribute specifies how many columns a cell should span (merges cells horizontally), while the `rowspan` attribute specifies how many rows a cell should span (merges cells vertically).

2.7.4 Merging Cells

Here's an example of a table with merged cells:

Code/Preview

blank-html-page.html

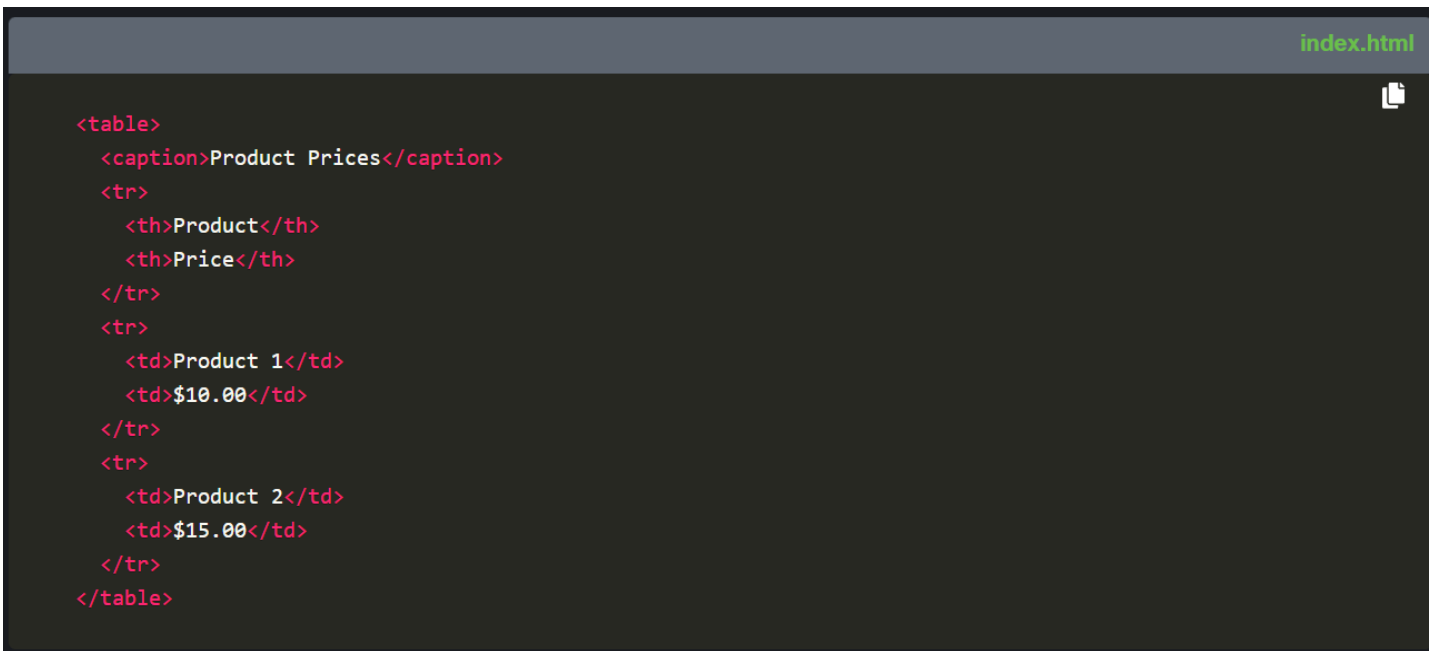
```
<table>
  <tr>
    <th rowspan="2">Product</th>
    <th colspan="2">Price</th>
  </tr>
  <tr>
    <th>Old</th>
    <th>New</th>
  </tr>
  <tr>
    <td>Product 1</td>
    <td>$10.00</td>
    <td>$12.00</td>
  </tr>
  <tr>
    <td>Product 2</td>
    <td>$15.00</td>
    <td>$18.00</td>
  </tr>
</table>
```

Product	Price	
	Old	New
Product 1	\$10.00	\$12.00
Product 2	\$15.00	\$18.00

2.7.5 The <caption> Element

The <caption> element is used to add a title or caption to an HTML table. It is a container tag that should be placed immediately after the opening <table> tag and before any other table elements.

For example, let's say you have a table that shows the prices of different products. You can use the <caption> element to add a title for the table, such as "Product Prices":

A screenshot of a code editor window titled "index.html". The editor has a dark background with light-colored text. It displays HTML code for a table. The code starts with an opening <table> tag, followed by a <caption> tag with the text "Product Prices". Then, there are two table rows. The first row has two header cells: "Product" and "Price". The second row has two data cells: "Product 1" and "\$10.00". The third row has two data cells: "Product 2" and "\$15.00". The table is closed with a </table> tag. A small icon of a document is visible in the top right corner of the editor window.

```
<table>
  <caption>Product Prices</caption>
  <tr>
    <th>Product</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Product 1</td>
    <td>$10.00</td>
  </tr>
  <tr>
    <td>Product 2</td>
    <td>$15.00</td>
  </tr>
</table>
```

2.7.6 The <table> Tag and Its Attributes

As you've learned, the HTML <table> tag helps you make tables on a webpage. But what if you want to change how the table looks? Well, you've got a couple of options: you can tweak the attributes of the <table> tag itself, or you can use CSS styles to jazz up the table.

In this course, we'll check out both ways to help you make tables that work well and look good. But for now, we're going to focus on playing with the attributes of the <table> tag. We'll save CSS for the CSS Essentials course, which is part 2 of the series.

The <table> tag in HTML has some attributes you can use to change how tables appear. You can add these attributes to the opening <table> tag, and they let you control things like the size and layout of the table, as well as the color and border style of the cells.

2.7.6 The <table> Tag and Its Attributes

Some of the most commonly used attributes include:

- ***border***, which sets the border size of the table (deprecated in HTML 5, CSS should be used instead).
- ***cellspacing***, which sets the space between cells in the table (deprecated in HTML 5, CSS should be used instead).
- ***cellpadding***, which sets the space between the cell content and the cell border (deprecated in HTML 5, CSS should be used instead).
- ***width***, which sets the width of the table (deprecated in HTML 5, CSS should be used instead).
- ***height***, which sets the height of the table.
- ***align***, which sets the horizontal alignment of the table (deprecated in HTML 5, CSS should be used instead).
- ***bgcolor***, which sets the background color of the table (deprecated in HTML 5, CSS should be used instead).

2.7.6 The <table> Tag and Its Attributes

Code/Preview

```
blank-html1-page.html

<table border="1" cellspacing="10" cellpadding="5" width="80%"
      height="200" align="center" bgcolor="#f2f2f2">
  <tr>
    <th>Product</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Product 1</td>
    <td>$10.00</td>
  </tr>
  <tr>
    <td>Product 2</td>
    <td>$15.00</td>
  </tr>
</table>
```

Product	Price
Product 1	\$10.00
Product 2	\$15.00